Revitalizing the Power of Blockchain Implementation for Transaction Processes in Real Estate Sector

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***Abstract*— Real estate is one of the most profitable assets in the world because it has advantages or significant impacts in the broader community, such as financial security, capable investment opportunities, economic stability, and growing the economy by empowering welfare and individual or corporate ownership through its transaction activities. But behind all these advantages, real estate transactions often face challenges such as high costs, lack of transparency and dependence on third parties. These problems will spread to other aspects, such as inefficient transactions because of the many entities that want to contribute, document falsification due to lack of transparency and disputes that can damage trust between stakeholders. Therefore, maintaining the security and validity of transaction documents is very much needed to minimize existing problems. To overcome these problems, applying blockchain technology in real estate transactions was proposed to create a safer, more transparent and efficient system. With blockchain's decentralized and immutable nature, it can overcome dependence on third parties and the problem of fraud risk. Supported by smart contracts and tokenization, blockchain enables the automation of transaction processes, thereby reducing costs, increasing speed, and strengthening trust at every transaction stage which can be more efficient and reliable.**

Keywords—Blockchain, Real Estate, Smart Contract, Tokenization, Transaction Process

I. INTRODUCTION

Real estate is one of the most sought-after assets worldwide and is the leading choice for investment because it covers various activities, from development to sales [1]. Owning real estate assets can provide long-term protection benefits from wealth and finances. Real estate assets can be essential in maintaining economic stability for individuals and companies because the price of these assets tends to be stable and can even increase over time [2]. However, despite having many advantages real estate asset transactions still need to overcome many challenges that hinder efficiency and trust. The high costs of conducting real estate transactions are caused mainly by fluctuations in land prices that vary significantly depending on the location of the land. Transactions can become increasingly complicated by the presence of third parties or the involvement of many intermediaries claiming to be stakeholders. Then the most common problem is the need for more transparency because document recording is still manual. An example is the falsification of ownership documents by showing the property price cheaply, but when the payment is completed, the approved letters are fake, and the property does not exist. This case can cause problems to spread to various aspects, such as distrust and harm to the economic problems of stakeholders. Therefore, an integrated system is needed to minimize various problems in real estate transactions so that they do not worsen in the future.

To overcome this problem, blockchain technology comes with an innovative solution. Blockchain is often called a distributed ledger because it can integrate the recording process in real time and ensure that data cannot be manipulated [3]. The nature of blockchain itself is that it is a decentralized system, which is data cannot be stored in one place but is spread across the network. In a decentralized system, each transaction will be verified by a network of independent nodes, so the network must approve each process or data change. Therefore, problems such as lack of transparency and potential manipulation will be reduced because no one party can control other ownership data.

Smart contracts are essential in implementing blockchain technology to overcome real estate sector transaction challenges [4]. Smart contracts can automate various transaction aspects from ownership verification to real-time payment processing. With smart contracts dependence on intermediaries will be reduced, reducing the risk of fraudulent activities [5]. For example, the automation process of ownership verification can ensure that only authenticated parties can carry out the transaction process, thereby reducing the risk of disputes.

Tokenization can implement blockchain in fractional ownership of real estate assets. By dividing ownership into various digital tokens, blockchain technology can increase liquidity value and can allow many investors who want to participate in real estate transactions [6]. Tokenization opens new opportunities for long-term shared ownership, making real estate investment more accessible to the broader community [7]. For example, fractional ownership through tokenization can make or help small investors own a portion of a high-value property that was previously unattainable.

This article explores how blockchain technology can minimize various challenges by analyzing its potential in real estate transactions. Through in-depth analysis and artifacts, it will show how blockchain features such as smart contracts and tokenization solve previously existing problems.

II. LITERATURE REVIEW

A. Real Estate

In real estate, blockchain technology provides a powerful solution to address longstanding challenges such as

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inefficiency, fraud, and lack of transparency. To successfully implement a blockchain-based real estate system, all key stakeholders—buyers, sellers, banks, insurance agencies, local authorities, and public registries—must collaborate and adopt the technology at each stage of the process [8]. Blockchain acts as a single source of truth, ensuring that property ownership records are secure and tamper-proof. This guarantees that ownership information is accurate and easily accessible, which streamlines the process of title transfer and reduces the risk of fraud. The digital identification of property owners plays a crucial role in this system, enhancing security and ensuring that ownership is verified before any transaction takes place [9]. Additionally, the integration of blockchain with smart contracts automates and secures transactions reducing the need for intermediaries and making the overall process faster and more efficient.

B. Blockchain Mechanism

Along with the development of the internet, various digital currencies such as Bitcoin and Ether have emerged. This makes blockchain technology increasingly popular as a significant driver in terms of financial progress. On the other hand, the property industry is one of the most promising sectors if it can utilize the great potential of blockchain technology. By having a verification method and secure data storage capabilities, blockchain is a secure tool for guaranteed data storage and simplifying property transactions. One of the advantages of the blockchain feature is its immutability, which cannot be changed. Later, if a transaction needs to be added, each node must validate each digital ledger. If it has been verified, a new transaction will be added to the ledger. Therefore, it can be said that this feature can increase transparency and prevent data fraud [10].

Technically, the process of forming a blockchain begins with the creation of an encryption code that can be called a "Hash". This code can be obtained through a cryptographic algorithm to be able to produce a value that represents data in each block. A block header containing a timestamp, a hash of the previous block and other important data will be formed. If you want to add a new block to the blockchain, a cryptographic mechanism called a nonce is needed[11]. Nonce has a function to ensure that each block has met the specified validation requirements. If the validation process is successful a new block will be added to the previous block chain, which aims to maintain the integrity and security of the relationship between blocks. This process can ensure that any data changes can be easily detected. When talking about the real estate sector, this mechanism can help ensure authenticity, transparency and the transfer process between property ownership. This process ensures that any data changes can be easily detected. In the real estate sector, this mechanism can help optimize the authenticity, transparency, and transfer of property ownership.

C. Smart Contract

Smart contracts can be defined as a system of rules that can be run as a program that can verify itself and cannot be manipulated on blockchain technology [12]. Conceptually, smart contracts act as protectors or guards between computerized transactions that act as validators in a contract. If blockchain technology and intelligent contacts are combined, they can provide a more flexible solution to overcoming various problems in the real estate industry.

Smart contracts have properties that are very much needed for this industry because this technology is immutable and traceable. Smart contract activities themselves can be managed publicly in a trusted environment in a safe manner and can enable peer-to-peer transactions [13]. An example is when a property owner (A) wants to sell his property to a buyer (B) without using a third party or intermediary such as a notary or broker. Then, they use the innovative contract system on the blockchain platform, which aims to ensure that transactions run automatically and transparently.

D. Related Works

Many researchers have researched various aspects of implementing blockchain technology in real estate. Based on Table 1, various headlines were classified, namely from the names of authors, key findings, and weaknesses they wrote in their papers. This table helps us obtain valid data and identify things that must be considered before writing this research.

TABLE I. KEY FINDINGS AND LIMITATIONS

Authors	***Key Findings***	***Limitations***
Sobhan Latifi et al.	Proposed a blockchainbased real estate market system using smart contracts to improve liquidity and transparency in commercial real estate.	High complexity of implementation and resistance from traditional market players hinder adoption.
Evgeny Pankratov et al.	Explored blockchain's potential to expedite real estate transactions, reduce fraud and lower costs through improved security.	Limited empirical examples in the Russian context and underdeveloped regulatory frameworks .
Kyaw Soe Lin et al.	Developed an adaptive model for blockchain in the Myanmar mortgage industry, enhancing validation processes and security.	Trust and infrastructure challenges in transitioning to a decentralized model; regulatory readiness issues.
Anniina Saari et al.	Conducted a systematic review of blockchain applications in land administration and real estate, which noting hybrid models enhance transparency.	Real-world scalability remains limited; hybrid applications do not fully realize blockchain's transformative potential.
Hugo Pieter Wouda & Raymond Opdenakker	Proposed a blockchain application for office building transactions in the Netherlands to enhance transparency and trust.	High initial development costs and the need for robust infrastructure for implementation.
Edvard Tijan et al.	Explored the potential of blockchain in logistics to enhance transparency and improve supply chain efficiency.	Identified scalability and performance challenges as well as high energy consumption issues.

III. METHODOLOGY

TABLE II. DESIGN SCIENCE RESEARCH METHODOLOGY

Headlines	***Descriptions***
Identify Problems and Motivate	Identification of problems in the transaction process in the real estate sector.
Define Objectives of Solution	Analysis of existing approaches to solutions: Smart contracts and tokenization in the property asset ownership transaction process.
Design and Development	Design of the blockchain modeling process of property assets.
Demonstration	Observe how effective and efficient the solution.
Evaluation	High initial development costs and the need for robust infrastructure for implementation.
Communication	Scholarly Publications.

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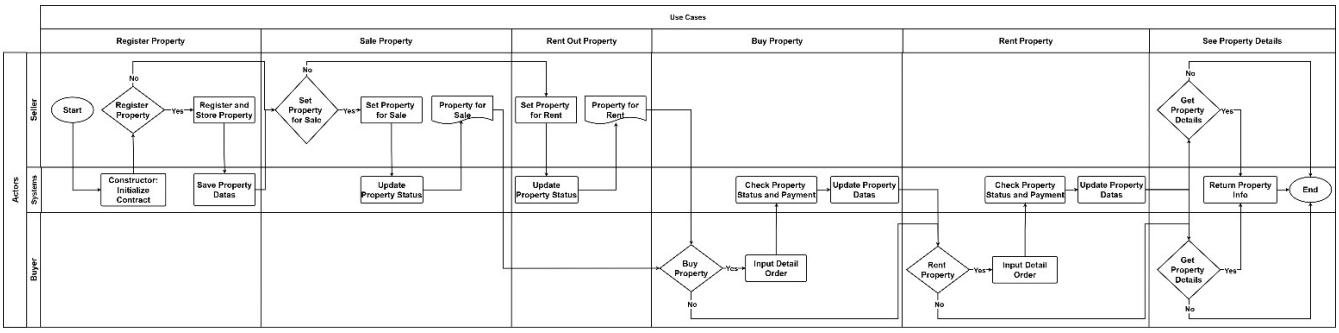


Fig. 1. Flowchart of the Blockchain-Based Real Estate System.

Design science research methodology is founded on pragmatic inquiry and strives to produce innovative 'artifacts' to resolve tangible real-world challenges [14]. Therefore, it holds significant relevance within the realm of information system research. As such, this approach is compatible and utilized to address field-specific challenges in this research. Table 2 shows the steps and processes underlying the design science research method.

IV. RESULT

In this section, several steps are designed to achieve the best possible results, starting with creating a flowchart-based model, system architecture, and the implementation of the smart contract system.

A. Model of Transaction Process (Flow Chart)

This flowchart illustrates how smart contracts play a role in each use case through the functions that have been developed. Smart contracts play a role in storing property registration data, validating property status (for rent or sale) and property transactions, and modifying ownership data of a property. With smart contracts, transparency can be increased, and fraud can be minimized in this property transaction because both buyers and sellers can see the ownership history of a property. For example, when a buyer wants to see ownership data of a property from year to year or when a seller wants to know the holders of the property he registered along with their respective ownership percentages.

Speaking of holders, this is where tokenization plays a role in our system. Buyers can make partial property purchases by purchasing several tokens available on a property with a forsale status. For example, a property worth $ 10 million can be broken down into 1 million tokens, with each token worth $ 10. With this tokenization system, the problem of property being viewed as an investment instrument requiring large funds can be addressed. This way, investors can buy tokens according to their ability, allowing them to own part of the property.

B. Blockchain-Enabled Property Transaction System: Data Access and Service Workflow.

In this section, a system architecture was designed for implementing smart contracts in blockchain-based property transactions. The system architecture is essential as it provides a structured approach to understanding the different components of the blockchain ecosystem [15]. It helps us identify potential bottlenecks, ensure system scalability, and guarantee transaction security and efficiency. The relationships among the three layers were integrated.

In the Data Access layer, the interactions between the smart contract, smart chain, and tokens are explained. The smart contract acts as the core, where all transactions are verified and recorded on the blockchain, while user data is processed through decentralized applications (DApps). Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

In the Service layer, the transaction verification process by blockchain network nodes is detailed. Nodes collaborate to verify and mine transactions, with new blocks being added to the chain once transactions are confirmed.

In the user layer, two actors are depicted engaging in a transaction using their electronic devices, which are already connected to the DApps server.

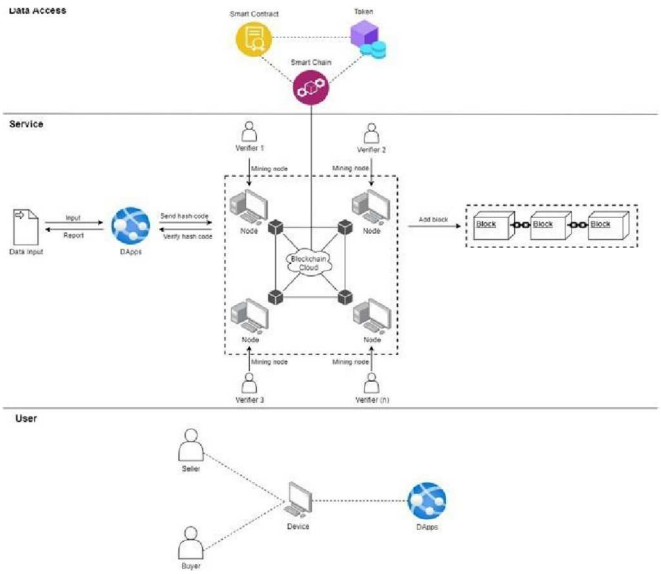


Fig. 2. System Architecture.

C. Smart Contract Implementation.

After designing the system flowchart and architecture, a smart contract-based system was developed and implemented with various functions to manage property transactions transparently and efficiently. To access these functions, several variables are used to store important information about listed properties and transactions made. By utilizing these functions, users can easily list, buy, and rent properties based on tokenized assets. In addition, a modifier is also added to limit access to changing the status of the property. With this modifier, only the owner who registers the property can change the status of the property. For example, when person A registers his property, then people B, C, and D also buy some tokens from the property to become part of the property's shareholders. When all the property tokens have been sold and the property shareholders agree to change the status of the property to open for rent, then the only person who can change the status of the property is person A.

TABLE III. DESCRIPTION, FUNCTIONS, AND RESULTS OF SMART CONTRACT

Description	*Functions*	*Result*

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Key Variable	properties, propertiesToken, propertiesCount, and totalTransaction	Stores complete information about registered properties.
Key Functions	registerProperty()	Register a new property and mint the ERC721 token.
	saleProperty()	List a property for sale.
	buyProperty()	Buy a property with tokens.
	rentOutProperty()	Set a property for rent.
	rentProperty()	Rent a property and pay the owner.
	purchaseTokens()	Purchases property tokens according to the desired amount.
	getProperty()	Retrieve property details.
Modifiers	onlyOwner(_id))	Restrict access to property owner.
Events	PropertyRegistered, PropertySold, PropertyRented, TokensPurchased, and BuyProperty	Emit events for property registration, sales, rentals, and token purchases.

D. Testing Smart Contract in Sepolia Testnet

After designing the smart contract, which includes various components such as variables, use cases, functions, and other requirements, it has been deployed to our system on a testnet for testing and validation. The smart contract was wusing the Remix IDE, an online integrated development environment for Ethereum smart contracts. After that, the system was deployed on Ethereum's test network (testnet), called the Sepolia Testnet. Before deploying on this platform, the necessary libraries for tokenization were imported, specifically the ERC- 721 standard library, which is used for creating non-fungible tokens (NFTs) [16]. Additionally, for the deployment and transaction process, a MetaMask wallet was set up to connect to the Sepolia testnet. Once everything was configured correctly, the deployment proceeded.

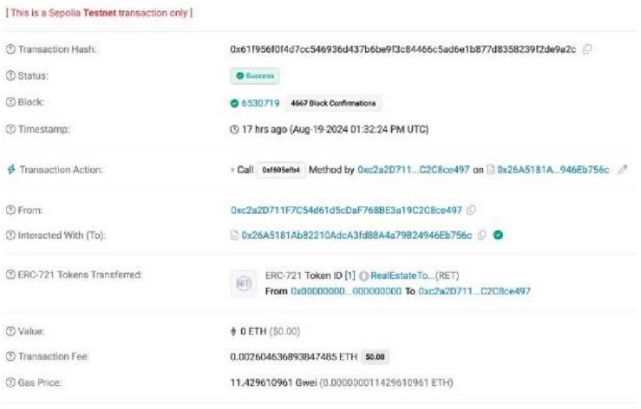


Fig. 3. Receipt Transaction of Register Property.

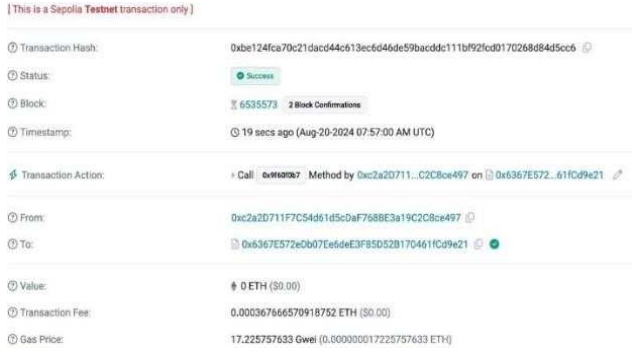


Fig. 4. Receipt Transaction of Sell Property.

At this stage of development, metric-based calculations are being compiled to access how technology impacts real estate transactions. As a sample, transaction data that includes the initial process namely property registration and sales is analyzed. After successful implementation, the system will direct to the Sepolia testnet website page containing transaction receipts. Figure 3 shows receipts from property registration activities, while Figure 4 shows receipts from property sales activities. Based on these two proofs of receipt, it can be conducted to an analysis that can be used as calculation metrics and validation, which will cover three main aspects: **cost, time**, and **fraud reduction**. This third aspect is the leading indicator in our assessment of the effectiveness of blockchain in real estate transactions. Later, the metric will be calculated using the data that has been predicted regarding the current situation in the real world.

a) Cost Efficiency Ratio

Existing real estate transactions are still carried out traditionally, which requires high costs because they involve many third parties such as brokers, legal advisors or administrative entities [17]. This process still requires various activities that will incur additional costs such as registration fees, sales commissions and other operational costs. Transaction costs on the blockchain come from gas fees, which are the computing costs required to run operations on the Ethereum network. In implementing our system, gas fees are analogized for property registration and sales transactions. These costs will later be compared with the traditional costs that has been assumed like usually incurred in real estate transactions. Based on **cost** calculation case, the gas fee required for property registration transactions is 0.000206 ETH. Assuming the price of Ethereum is **$ 2,000 per ETH**. So, the total is USD 0.41. Meanwhile, the gas fee required for property sales is slightly higher, 0.000368 ETH. If calculated, the cost is USD 0.73. The total cost of blockchain technology is 0.000574 ETH or **1.14 USD**. Therefore, to find the ratio value of its cost efficiency, the existing costs must be compared with the total blockchain costs that have been previously calculated.

$$Cost\ Efficiency\ Ratio = \frac{Traditional\ Cost}{Blockchain\ Cost} = \frac{2000}{1.14} = 1754x \quad (1)$$

b) Time Efficiency Ratio

Traditional real estate transactions take a long time because they require various parties, such as legal advisors or brokers. Of course, this process will take many stages, such as property search, negotiation, and signing the sale and purchase transaction [18]. The legal document examination stage in manual transactions can take 1 to 2 weeks, and the financing process through a bank intermediary can take 2 to 4 weeks. Overall, the total duration of traditional transactions can take 6 to 12 weeks depending on the complexity and efficiency of each stage [19] [20]. In this formula, property registration are assumed to have an average time of 5 days (432,000 seconds), while property sales activities are 7 days (604,800 seconds). Based on the testing results on the testnet, the time needed for registration is around 30 seconds, and the duration of registration to sale is 37 minutes (2,209 seconds). With this data, it can be assume with the following formula.

$$Regist\ Property = \frac{Traditional\ Time}{Blockchain\ Time} = \frac{432.000}{30} = 14.400x \quad (2)$$

$$Traditional\ Time\ 604.800$$

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$$Sell\ Property = \frac{}{} = = 273.79x \quad (3)$$

$$Blockchain\ Time \quad 2209$$

c) Fraud Risk

Traditional real estate transactions are still very vulnerable to fraud due to the heavy reliance on various intermediaries and lack of transparency. In 2017, the Fuderal Bureau of Investigation (FBI) estimated that fraud in the US reached more than **$1 billion**. The National Association of Realtors (NAR) revealed that 10% of total real estate transactions are affected by fraud, which still often occurs with the excuse of title fraud and identity theft.

In the other hand, blockchain technology significantly reduces the risk of fraud in real estate transactions by providing a transparent, immutable and decentralized ledger system. Here are some key points that support how blockchain improves security and reduces the potential for fraud:

- Enhanced Transparency and Immutable Records

Blockchain transparency enables increased security in real estate transactions, reducing fraud and growing trust. The study claims that blockchain can reduce fraud by 90-95% compared to traditional systems, making it a highly secure solution [21].

- Industry-wide Fraud Reduction

A study by PricewaterhouseCoopers (PWC) highlights that implementing blockchain in real estate can reduce transaction costs while increasing security, potentially saving **$1.1 billion annually** in fraud-related losses in the real estate sector. This reduction directly results from blockchain's ability to securely store and verify transaction records without needing a third party.

- Reduced Documentation Errors

Blockchain can reduce document errors, which are typical in traditional real estate transactions. This reduction in errors directly contributes to reduced fraud, as there is less opportunity for documents to be forged or misrepresented during the transaction process [22].

- Reduced Reliance on Third Parties

Accenture (2020) states that blockchain reduces the need for third-party intermediaries in real estate. By eliminating unnecessary third parties, the fraud is reduced as there is less opportunity for manipulation or falsification of transaction data.

d) Metrics Summary

Our analysis revealed that blockchain technology can make significant improvements in reducing the risk of fraud, as well as in time efficiency and cost efficiency. All improvement aspects are summarized in a table, which compares traditional real estate systems and blockchain-based systems.

TABLE IV. METRICS SUMMARY OF EFFICIENCY

Metric	***Traditional Systems***	***Blockchain***	***Efficiency***
Cost Efficiency	Registration: $500 - $2,000 USD Additional costs: Legal fees, closing costs	Gas fee (registration: 0.000206 ETH, sale: 0.000368 ETH)	$1.14 USD total cost
Time Efficiency	6-12 weeks (432,000 - 604,800 seconds)	Registration: 30 seconds; Sale: 2209 seconds (~37 minutes)	14,400 times faster (registration) & 273.79 times faster (sale)
Fraud Risk	50 fraud incidents per year (Fuderal Bureau of Investigation (FBI) and National Association of Realtors (NAR) estimate)	5-10 fraud incidents per year (Blockchain reduces fraud by 90-95%)	10 times lower fraud risk (based on 90-95% fraud reduction)

V. DISCUSSION

Based on our research in implementing blockchain technology, optimal results have been achieved. This is supported by formulas (1), (2), and (3) which present a very significant comparison to the current manual transaction process. The results of our research also provide significant solutions in facing challenges in real-estate transactions. By using metric analysis, concrete improvements in terms of time efficiency, costs, and lack of transparency are well observed.

Furthermore, with the presence of smart contracts, the transaction process will be faster and more efficient, which is followed by reducing costs and dependence on third parties. Additionally, the tokenization discussed in this paper enables small-scale investors to participate, opening more inclusive investment opportunities and increasing the liquidity of real estate assets. However, of course there are still challenges in implementing this blockchain, including market resistance to adopting new technologies like blockchain, the need for regulatory frameworks supporting blockchain applications in real estate transactions, and issues related to the sensitivity of transactional data.

To address these challenges, these are solutions that can be implemented to facilitate blockchain technology in real estate transactions. The problem of market resistance can be solved by holding an educational campaign that contains the real benefits of blockchain technology for stakeholders to help create trust and further understanding. Regulatory framework problems can be resolved by collaborating with policymakers who will later develop structured and clear guidelines for integrating blockchain. The problem of transaction data sensitivity can be solved using advanced encryption techniques such as secure enclaves or data masking, which can later ensure confidentiality without eliminating transparency. By implementing these various solutions, the problems faced in adopting blockchain in the real estate sector can be minimized effectively.

Various solutions owned by blockchain technology in overcoming its problems make blockchain a new, innovative, and revolutionary technology. Having not being able to be manipulated and transparent ensures that the parties involved will feel safe in terms of trust. Overall, the results of our study confirm and convince us that blockchain technology has the potential to be the foundation of the future, especially in the real estate sector.

VI. CONCLUSION

The results of this paper can be concluded that the implementation of blockchain technology has been proven to significantly enhance efficiency in real estate transactions, as demonstrated by the three formulas presented in this paper. Furthermore, this paper also shows that the smart contract implemented in the system can successfully automated various property transaction processes from registration to sales, by minimizing dependence on third parties and increasing transaction transparency. Additionally, the

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tokenization that has been discussed in this paper also allows partial ownership of property which can open investment opportunities for small-scale investors and increase the liquidity of real-estate assets.

To develop this system in the future, several suggestions can be optimized in implementing blockchain in the real estate sector, such as integrating with the government to create a legal and certified application regarding land management, implementing a multi-signature system and recovery mechanism to enhance transaction security and protect user's digital assets and an integrated marketplace feature will be added to allow users to search, compare, and conduct property transactions directly through the platform.

By implementing these systems, blockchain technology can effectively address various challenges in real estate transactions, as previously mentioned. This research also proves that blockchain technology has the potential to revolutionize the real estate industry, particularly with further development in the areas highlighted above.

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